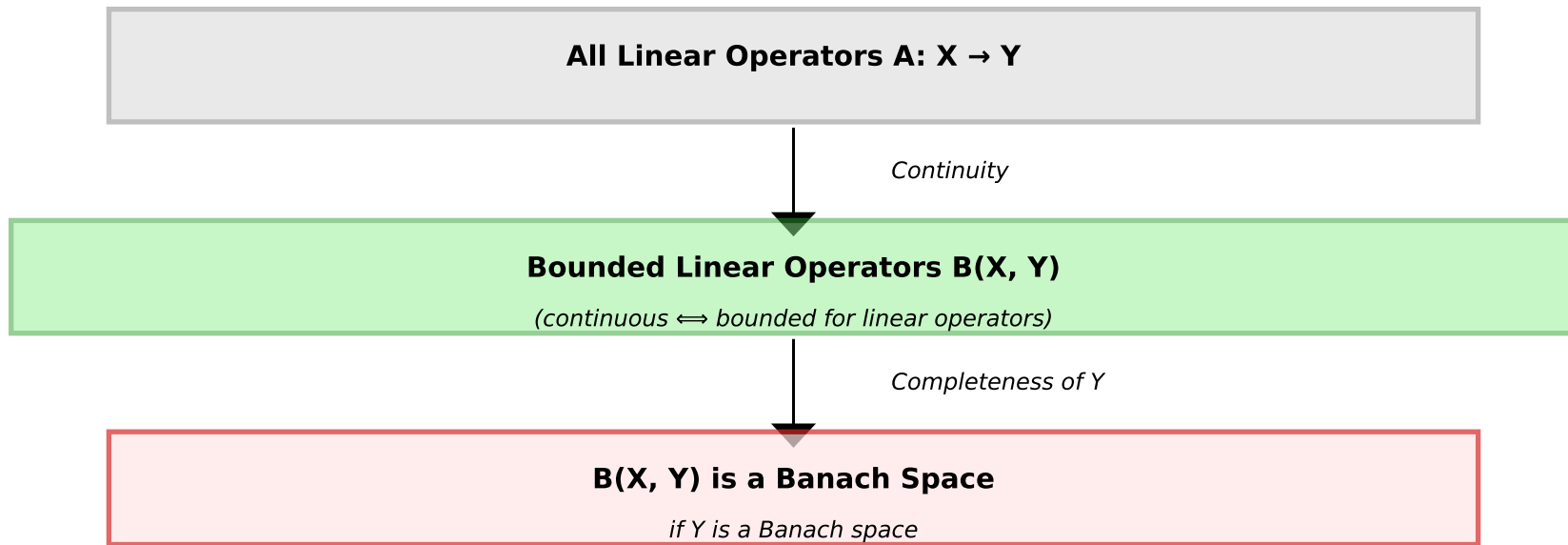


Hierarchy of Operator Spaces



Key Properties of $B(X, Y)$:

- Norm: $\|A\|_{B(X,Y)} = \sup \{ \|A(x)\|_Y : \|x\|_X \leq 1 \}$
- Linear space under operator addition and scalar multiplication
- Normed space with complete metric if Y is Banach
- Contains identity operator I and zero operator 0
- Closed under composition (if codomain = domain)

Examples:

- $B(\mathbb{R}^n, \mathbb{R}^m)$: $n \times m$ matrices with operator norm = largest singular value
- $B(C[a,b], C[a,b])$: multiplication by continuous function bounded by $\max|f|$
- $B(L^2(\Omega), L^2(\Omega))$: integral operators with finite operator norm